ECON 0100 | Exercise C2 | Taxes & Subsidies

Solutions. Answers and work are shown in the red blocks; everything else is what students see.

In recent years Hogsmeade has seen a proliferation of candy shops selling smoked toffees. The Demand (marginal benefit) curve and Supply (marginal cost) curve for smoked toffee can be represented by the following relationships.

$$D : P_B = 20 - \frac{1}{2}Q_D \qquad\qquad S : P_S = 2 + Q_S$$

Due to the metabolic issues among the younger generation of wizards in the community resulting from the consumption of these high glycemic index treats, the Ministry of Magic imposed a 3 Galleon tax on the sale of smoked toffee.

Q1 | One Toffee Exchange

First Exchange: A buyer values one smoked toffee at 16 Galleons. It costs the seller 12 Galleons to make.

a) Without the tax, will the buyer and seller make this exchange? b) With the tax, will the buyer and seller make this exchange? c) Total Surplus to the buyer and seller, with the tax: d) Tax paid to the Ministry of Magic: e) Total Surplus, including the Ministry of Magic:

a) Yes. **b)** Yes. **c)** 1 Galleon. **d)** 3 Galleons. **e)** 4 Galleons.

a) The buyer's value of 16 is above the seller's cost of 12, so there is 4 of surplus to split and the trade happens.

b) With the tax, the buyer's 16 has to cover the seller's cost *and* the tax: $12 + 3 = 15 \leq 16$, so the trade still happens. Whichever side hands the tax to the Ministry, the buyer pays at most 16 and the seller keeps at least 12.

c) $16 - 12 - 3 = 1$ Galleon is left for the buyer and seller to share.

d) 3 Galleons.

e) $1 + 3 = 4$ Galleons, the same as without the tax. On a trade that still happens, the tax moves surplus from the buyer and seller to the Ministry; it does not destroy any.

Another buyer values one smoked toffee at 14 Galleons. It costs the seller 12 Galleons to make.

f) Without the tax, will the buyer and seller make this exchange? g) With the tax, will the buyer and seller make this exchange? h) Deadweight Loss from losing this exchange:

f) Yes. **g)** No. **h)** 2 Galleons.

f) $14 > 12$, so the trade creates 2 of surplus and happens.

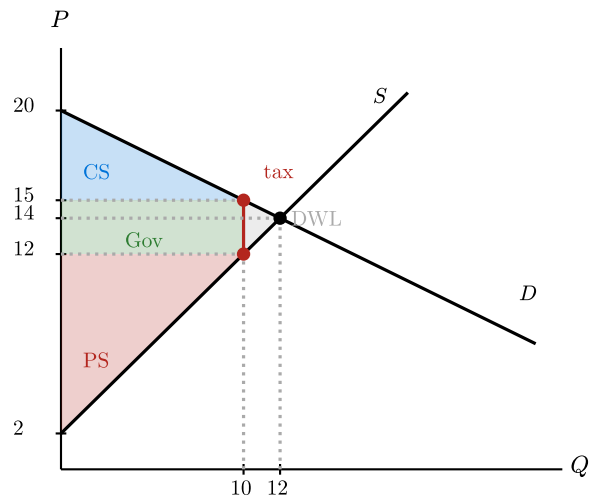
g) The buyer's 14 cannot cover $12 + 3 = 15$, so no price works for both sides and the trade does not happen.

h) The 2 of surplus this trade would have created is lost to everyone. The Ministry collects nothing on a trade that never happens, so nothing offsets the loss: this is deadweight loss.

Q2 | Tax Equilibrium

Use a graph and algebra to discuss the impact of the tax on the market. Include how the welfare areas changed after the policy. No need to calculate welfare measures except DWL.

Pre-Tax Price: 14. Pre-Tax Quantity: 12. Post-Tax Quantity: 10. DWL: 3. Buyer Price: 15. Seller Price: 12. Tax Incidence on Buyers: 1. Tax Incidence on Sellers: 2. Who bears more of the tax? Sellers.



Without the tax, buyers and sellers face one price:

$$20 - \tfrac{1}{2}Q = 2 + Q \quad \Rightarrow \quad Q = 12, \quad P = 14$$

With the tax, the buyer price sits 3 above the seller price, and the market still clears at one quantity. Slide the 3-Galleon wedge in from the left until it gets stuck between the curves:

$$P_B = P_S + 3 \quad \Rightarrow \quad 20 - \tfrac{1}{2}Q = (2 + Q) + 3 \quad \Rightarrow \quad Q = 10, \quad P_S = 12, \quad P_B = 15$$

Incidence: buyers pay $15 - 14 = 1$ more than before, sellers receive $14 - 12 = 2$ less. Sellers bear more because supply is steeper (less elastic) than demand: sellers are less able to walk away, so more of the tax lands on them.

Welfare: consumer surplus shrinks from the triangle above 14 to the triangle above 15 (36 to 25); producer surplus from the triangle below 14 to the one below 12 (72 to 50). The rectangle between the two prices over the 10 toffees still sold is tax revenue, $3 \times 10 = 30$. The triangle between the curves over the 2 toffees no longer sold is deadweight loss:

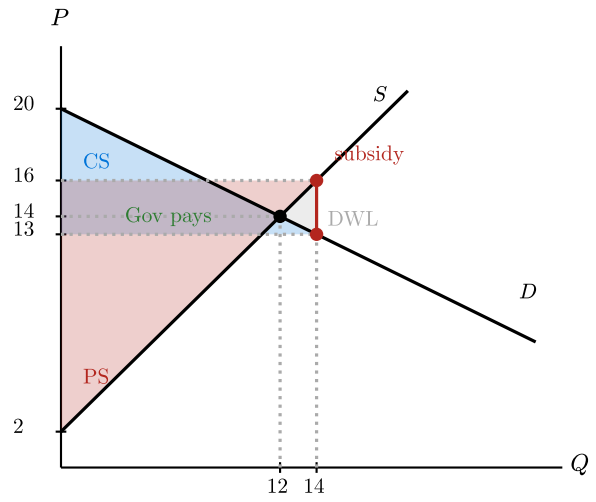
$$\text{DWL} = \tfrac{1}{2} \times 3 \times (12 - 10) = 3$$

Check: $36 + 72 = 25 + 50 + 30 + 3$.

Q3 | Subsidy Equilibrium

Now suppose that instead of the tax, the Ministry of Magic pays a 3 Galleon subsidy on every smoked toffee sold. Use a graph and algebra to discuss the impact of the subsidy on the market.

Post-Subsidy Quantity: 14. DWL: 3. Buyer Price: 13. Seller Price: 16.



A subsidy is the tax wedge turned around: the seller now receives 3 more than the buyer pays, so the wedge sits to the *right* of the no-policy equilibrium:

$$P_S = P_B + 3 \quad \Rightarrow \quad 2 + Q = (20 - \tfrac{1}{2}Q) + 3 \quad \Rightarrow \quad Q = 14, \quad P_B = 13, \quad P_S = 16$$

Both sides are better off: buyers pay 1 less and sellers receive 2 more, so consumer surplus rises to 49 and producer surplus to 98. But the Ministry pays $3 \times 14 = 42$, more than the $(49 - 36) + (98 - 72) = 39$ the buyers and sellers gain. The difference is the deadweight loss, the triangle between the curves over the two added toffees. The 13th and 14th toffees cost more to make (15 and 16) than any buyer values them (13.5 and 13); they are made only because the Ministry pays the gap:

$$\text{DWL} = \tfrac{1}{2} \times 3 \times (14 - 12) = 3$$